

Big Data Analytics (ADT302) Question Bank Analysis

Module I: Introduction to Big Data

Part A Questions (3 marks each)

1. List the different characteristics of Big data. (Jan 2024)
2. What is Big data Platform? What are its characteristics? (Jan 2024)
3. What is data? Explain about nature of data. (June 2023)
4. Give examples of different Big Data platforms. (June 2023)
5. Explain the differences between data analysis and data reporting. (May 2024)
6. **What is big data platform? What are its features?** (May 2024) - **Repeated from Jan 2024**
7. **List out the 5 Vs of Big Data.** (Dec 2024) - **Similar to Jan 2024 Q1**
8. Write a note on any three big data analytics tools. (Dec 2024)

Part B Questions (14 marks each)

1. a) Explain in detail about Nature of Data and its types. (Jan 2024) b) Illustrate Big Data Architecture. (Jan 2024)
2. a) Explain in detail the benefits of Intelligent Data Analysis. (Jan 2024) b) Discuss the importance of Big Data. (Jan 2024)
3. a) Describe the various big data analytical methods in detail. (June 2023) b) Explain Intelligent Data Analysis in detail. (June 2023)
4. a) Define the term "Big data". Which are the different 5 V's that help to decide whether a given data source contributes to big data. (June 2023) b) Explain in detail about data analytic processes and tools. (June 2023)
5. a) Explain big data architecture in detail with the help of a neat diagram. (May 2024) b) What are the different phases in big data analytics life cycle? Explain in detail. (May 2024)
6. a) Explain the Evolution of Big Data and their characteristics. What are the different types of big data? (May 2024) b) Explain in detail about Nature of Data and its applications. What are the different types of data. (May 2024)
7. a) **Illustrate Big Data Architecture and explain each component in detail.** (Dec 2024) - **Repeated from Jan 2024/May 2024** b) Differentiate conventional data and big data. (Dec 2024)
8. a) **Recall and explain different phases in big data analytics life cycle?** (Dec 2024) - **Repeated from May 2024** b) Compare the types of Big Data with examples. (Dec 2024)

Module II: Stream Processing

Part A Questions (3 marks each)

1. What are the benefits of stream processing? (Jan 2024)
2. Illustrate the stream data architecture. (Jan 2024)
3. What are the key components of streaming data architecture? (June 2023)
4. Is conventional data processing sufficient for stream processing. Why? (June 2023)
5. What are the challenges in stream processing? (May 2024)
6. Describe estimating moments and its types. (May 2024)
7. **How does stream processing benefit over traditional data processing?** (Dec 2024) - **Similar to Jan 2024 Q3**
8. Some websites check availability of username by searching millions of usernames registered with it. Identify one effective method to filter data as in this type of scenario. (Dec 2024)

Part B Questions (14 marks each)

1. a) Explain Sampling techniques for efficient stream processing. (Jan 2024) b) Explain DGIM algorithm with an example. (Jan 2024)
2. a) Explain Bloom filter with an example. (Jan 2024) b) How to use Flajolet- Martin algorithm to count the distinct elements in a stream? (Jan 2024)
3. a) What is Bloom filter and what are its properties. Illustrate the working of Bloom filter with examples. (June 2023) b) Illustrate Flajolet- Martin algorithm. (June 2023)
4. a) Explain in detail the algorithm used for counting the number of ones in a window. Consider there is a window of length $N=24$ on a binary system. Find the total number of ones in the last 20 bits, when 0111 enters into the given stream101011000101110110010110.... The new bit enters from the right side and time stamp of first new bit is 100. (June 2023) b) Explain the term moments of the stream, and list out its importance. (June 2023)
5. a) Illustrate working of Bloom filter with examples. Explain in detail the operations that a Bloom filter supports. (May 2024) b) Find the number of distinct elements in the given input stream 1,3,2,1,2,3,4,3,1,2,3,1. The hash function is $h(x)=6x+1 \bmod 5$. Explain in detail the algorithm used to count distinct elements in the stream. (May 2024)
6. a) What are the different sampling techniques for efficient stream processing? Explain in detail. (May 2024) b) Illustrate DGIM algorithm in detail with example. Explain the rules for forming the buckets. (May 2024)
7. a) Describe Stream data architecture with an explanation and suitable illustration. (Dec 2024) b) Illustrate the working of Bloom filter with examples for i) Inserting an element ii) Searching an element. (Dec 2024)
8. a) What are the key challenges in Stream Processing? (Dec 2024) b) Suppose we have a window of length N (say $N=24$) on a binary system, we want at all times to be able to answer a query of the form "How many 1's are there in the last K bits?" for $K \leq N$. Suggest an algorithm to solve this issue with detailed explanation. Find the total number of ones, when 0111 enters into the given stream

...101011000101110110010110.... (Assume, the new bit enters from the right side and time stamp of first new bit is 100). (Dec 2024)

Module III: Hadoop Framework

Part A Questions (3 marks each)

1. What are the functions of datanode and namenode? (Jan 2024)
2. List the different Hadoop commands for basic file system operations. (Jan 2024)
3. Why do we need Hadoop for Big Data Analytics? (June 2023)
4. Give examples of Hadoop file systems. (June 2023)
5. **Compare name node and data node in HDFS.** (May 2024) - **Similar to Jan 2024 Q5**
6. Describe the Map Reduce job implementation in the case of Road Enrichment Example. (May 2024)
7. **Explain the functions of data node and name node.** (Dec 2024) - **Repeated from Jan 2024/May 2024**
8. What are the uses of Apache Hadoop? (Dec 2024)

Part B Questions (14 marks each)

1. a) Write about the anatomy of file write operation in HDFS. (Jan 2024) b) Describe Hadoop architecture with neat sketch. (Jan 2024)
2. a) Explain the Map reduce execution pipeline. (Jan 2024) b) Explain in detail Hadoop Ecosystem and its Core Components. (Jan 2024)
3. a) How wordcount problem can be implemented using MapReduce model? (June 2023) b) Write about the anatomy of file read operation in HDFS. (June 2023)
4. a) Explain map reduce programming model. (June 2023) b) Which are the core components of Hadoop? Explain in detail. (June 2023)
5. a) Illustrate map reduce job execution flow. (May 2024) b) Explain in detail the anatomy of file write operation in HDFS with neat diagram. (May 2024)
6. a) Explain how map reduce can be used as a framework for parallel processing. (May 2024) b) What is Hadoop ecosystem? Explain in detail its components with neat diagram. (May 2024)
7. a) Describe Hadoop ecosystem with neat sketch. (Dec 2024) b) **Explain in detail the anatomy of file read operation in HDFS with neat diagram.** (Dec 2024) - **Repeated from June 2023**
8. a) **How Can the Word Count Problem Be Implemented Using the MapReduce Model?** (Dec 2024) - **Repeated from June 2023**

Module IV: Pig, Hive and HBase

Part A Questions (3 marks each)

1. What are the various Execution Modes of Pig? (Jan 2024)
2. List the various DML commands supported by HBase. (Jan 2024)
3. **What are the different execution modes of Pig?** (June 2023) - **Repeated from Jan 2024**
4. Differentiate Hive from traditional databases. (June 2023)
5. State the differences between Pig and SQL. (May 2024)
6. **What is Hbase? What are its features?** (May 2024) - **Similar to Jan 2024 Q8**
7. **Identify the ways in which a pig program can be executed.** (Dec 2024) - **Similar to Jan 2024 Q7/June 2023 Q7**
8. Write note on Hive Metastore? (Dec 2024)

Part B Questions (14 marks each)

1. a) Explain the Architecture of Hive. (Jan 2024) b) Compare HBase with RDBMS. (Jan 2024)
2. a) What is HBase and what are its features? Explain HBase Data Model. (Jan 2024) b) List out the different data processing operations in Pig. (Jan 2024)
3. a) Explain the main components of the Hadoop Pig framework (June 2023) b) Write about different data processing operators in Pig Latin. (June 2023)
4. a) What are the different types of Tables in Hive? Explain. (June 2023) b) Explain the data model and Read-Write operations in HBase? (June 2023)
5. a) **Explain Pig architecture in detail. What are its components?** (May 2024) - **Similar to June 2023 Q17a** b) Illustrate features and Architecture of Hive with neat diagram. (May 2024)
6. a) Write the syntax to create a table and partition in Hive. (May 2024) b) What are the different data types used in Pig Latin and describe its various operators. (May 2024)
7. a) How Does Pig Compare to Hive? (Dec 2024) b) **Explain Pig architecture in detail. What are its components?** (Dec 2024) - **Repeated from May 2024**
8. a) **What is HBase and what are its features? Explain HBase Data Model.** (Dec 2024) - **Repeated from Jan 2024** b) **What are the different data types used in Pig Latin and describe its various operators.** (Dec 2024) - **Repeated from May 2024**

Module V: R Programming

Part A Questions (3 marks each)

1. Discuss the general list operations in R with example. (Jan 2024)
2. Discuss the features of R Language. (Jan 2024)
3. What are the important features of R programming? (June 2023)
4. Write an R program to find the factorial of a number by reading the input from keyboard. (June 2023)
5. Illustrate any three R functions used in data analytics. (May 2024)

6. Write an R program to print Fibonacci series. (May 2024)

7. **List out the key features of R Programming Language.** (Dec 2024) - **Similar to Jan 2024 Q10/June 2023 Q9**

8. Write an R program to find sum of two numbers. (Dec 2024)

Part B Questions (14 marks each)

1. a) Construct a data frame using R and perform different operations on it. (Jan 2024) b) Illustrate the operations performed in R vectors. (Jan 2024)
2. a) List the applications of R programming. (Jan 2024) b) Write an R program to find the factorial of a number by reading the input from a file and check whether it is palindrome. (Jan 2024)
3. a) Write a program in R to read a dataset 'internals.csv' which consists of id, name and marks of 5 subjects each out of 50. Write code i). To Read data frame and display columns ii). To add an extra column Total iii). To display class topper iv). To display student details having total marks above the average v). To prepare rank list by sorting rows based on total marks (June 2023) b) Discuss how reading and writing files is handled in R. (June 2023)
4. a) Explain Matrix creation and Matrix handling in R. Discuss about applying functions to Matrix rows and columns. (June 2023) b) Illustrate the operations performed in R vectors. (June 2023)
5. a) Describe about vectors in R. What are the operations that can be performed on vectors? (May 2024) b) Explain the different categories of attributes and data types in R. (May 2024)
6. a) What are lists in R? How lists are handled? (May 2024) b) Describe data frames in R. How it can be created and explain the various operations on frames. (May 2024)
7. a) Explain in detail different Matrix handling operations used in R Programming Language. (Dec 2024) b) List and explain three R functions used in descriptive statistics. (Dec 2024)
8. a) Discuss any four types of data visualization in R. (Dec 2024) b) Describe the advantages and disadvantages of data visualization in R language. (Dec 2024)